

What are Embeddings?

How a machine turns meaning into numbers — and compares them

TEXT → NUMBERS

COSINE SIMILARITY

MODEL CHOICE

1
2

What You'll Learn Today

1
2 3

What an embedding is

Meaning turned into numbers



Measuring closeness

Reading a similarity score



A real test

Five phrases, real scores, both models



Choosing a model

Cost, quality and language

1
2
3

What is an Embedding?

DEFINITION

A list of numbers that carries the meaning of a text

Similar meaning → similar numbers

FROM OUR OWN SYSTEM

"I want my money back"



embed

[-0.0114, -0.0205, -0.0579, 0.0246, ...]

1,536 numbers — always



No single number means anything on its own

The meaning lives in the position of the whole list



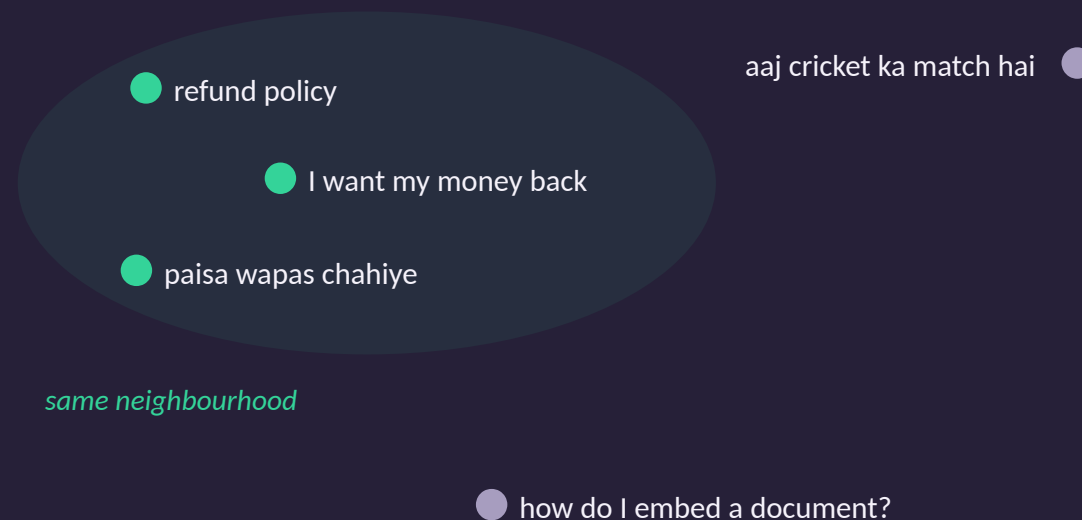
The Map of Karachi

EVERYDAY VERSION

- Every place has GPS coordinates
- Close places get close coordinates
- Saddar and Empress Market — almost the same
- Saddar and Port Qasim — nowhere near

Embeddings do the same — with meaning

THE MEANING MAP



Drawn here as 2 dimensions — really 1,536

One Vector Per Chunk — Not Per Document

INPUT

1 document

about 50 lines



SPLIT

5 chunks

about 10 lines each



EMBED

5 vectors

1,536 numbers each



Why not one vector per document?

A question about one line would have to match the meaning of all fifty. The match goes weak.



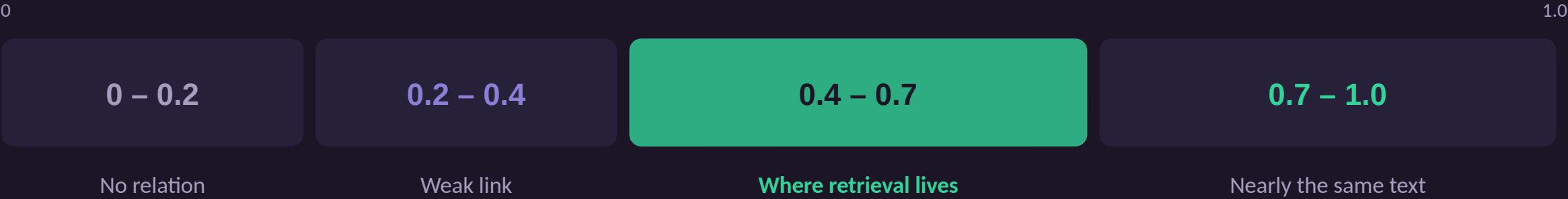
Who picks 1,536?

OpenAI did, when the model was built. It is a property of the model — not a setting you choose.



Reading a Cosine Score

How close do two vectors point in the same direction?



A score is not a percentage

0.451 does not mean "45% the same". A score only means something next to another score.

OpenAI recommends cosine — its vectors already come out at length 1



The Same Score, Two Verdicts

A score only means something next to the other scores in your own data.

ILLUSTRATION — NOT
MEASURED

NARROW DATA

One company · refund, returns, delivery

Refund policy	0.68
Return process	0.64
Delivery times	0.58
Office hours	0.49

0.49 — and still the wrong chunk

WIDE DATA

Our book · n8n, RAG, Instagram, deploy

Embeddings lesson	0.41
Vector database lesson	0.33
n8n webhook lesson	0.07
Instagram token lesson	0.05

0.41 — and clearly the right chunk

The smaller number won. A percentage could never do that.



Five Phrases, Real Scores

Embedding 5 phrases with both models...

text-embedding-3-small 5 phrases -> 1536 numbers each
text-embedding-3-large 5 phrases -> 3072 numbers each

Cosine similarity (higher = closer in meaning)

PAIR	3-small	3-large
"I want my money back" vs "refund policy" no shared words, yet the top score	0.451	0.453
"I want my money back" vs "paisa wapas chahiye" same meaning, different language	0.245	0.483
"paisa wapas chahiye" vs "aaj cricket ka match hai" same language, nothing else shared	0.273	0.272
"I want my money back" vs "how do I embed a document?" unrelated - the control	0.066	0.078

THE TEST - does meaning beat language?

text-embedding-3-small	meaning 0.245	language 0.273	-> LANGUAGE WINS (the problem)
text-embedding-3-large	meaning 0.483	language 0.272	-> <u>meaning wins</u>

0.451

No shared words

Not one word in common — and still the highest score on the board.

0.066

Unrelated stays near zero

The score is honest at both ends, not just the top.

Our own run — 7 August 2026, both models



When Language Beat Meaning

A higher score means the model thinks the two texts are closer.

SHOULD HAVE SCORED HIGHER

"I want my money back"
"paisa wapas chahiye"

Same meaning, different language

0.245

SHOULD HAVE SCORED LOWER

"paisa wapas chahiye"
"aaj cricket ka match hai"

No connection at all

0.272

BUT IT WON



The order came out backwards

Sharing a language counted for more than sharing a meaning

0.245 < 0.272

Measured on text-embedding-3-small — our own run, 4 August 2026



3-small vs 3-large

	text-embedding-3-small	text-embedding-3-large
Dimensions	1,536	3,072
Price / 1M tokens	\$0.02	\$0.13
English test (MTEB)	62.3%	64.6%
Multilingual test (MIRACL)	44.0%	54.9%
Storage per vector	1×	2×



English content and users

Take 3-small. The quality gap is small — the price gap is not.



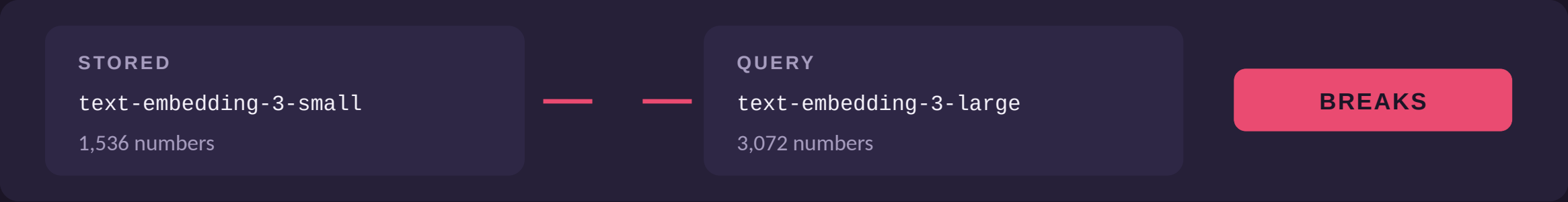
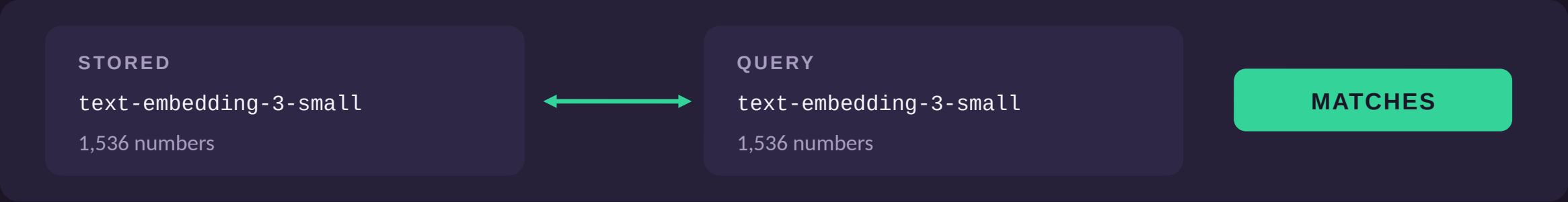
Mixed languages

3-large is the candidate. Your own queries decide it.



One Rule You Can't Break

Index and query must come from the same model — always.



Changing the model means re-embedding everything

So write down which model built a collection — six months from now you will need to know.

✓ Five Things to Remember

- 01 An embedding is meaning, written as numbers
- 02 Cosine compares direction, not words
- 03 A score is relative — never a percentage
- 04 A small model can miss another language
- 05 Same model for index and query — always

Next — Lesson 03: Vector Databases & Qdrant



Two Questions for You

01

A chunk scores 0.55. Is that a good match — or not?

02

Your users write Roman Urdu, your documents are English. What would you test first?

Answer both in the comments — the best one opens the next lesson.

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